

12 Diagnostics and Protection

12.1 Short Protection

The TMC2660C protects the MOSFET power stages against a short circuit or overload condition by monitoring the voltage drop in the high-side MOSFETs, as well as the voltage drop in sense resistor and low-side MOSFETs (Figure 12.1). A programmable short detection delay (*shortdelay*) allows adjusting the detector to work with different power stages and load conditions. Additionally, the short detection is protected against single events, e.g., caused by ESD discharges, by retrying three times before switching off the motor continuously. The low side short detector also provides for overcurrent detection. It needs to be explicitly enabled by programming.

Parameter	Description	Setting	Comment
SHRTSENS	The high-side overcurrent detector can be set to a higher sensitivity by setting this flag. This will allow detection of wrong cabling even with higher resistive motors.	0/1	0: Low sensitivity 1: High sensitivity
TS2G	Short detection delay for high-side and low side detectors. The short detection delay shall cover the bridge switching time. %01 will work for most applications. A higher delay makes detection less sensitive to capacitive load.	0/1	%00: 3.2µs. %01: 1.6µs. %10: 1.2µs. %11: 0.8µs.
DIS_S2G	Allows to disable short to VS protection.	0/1	Leave detection enabled for normal use (0).
EN_S2VS	Explicitly enable short to VS and overcurrent protection by setting this bit.	0/1	Enable detection for normal use (1).

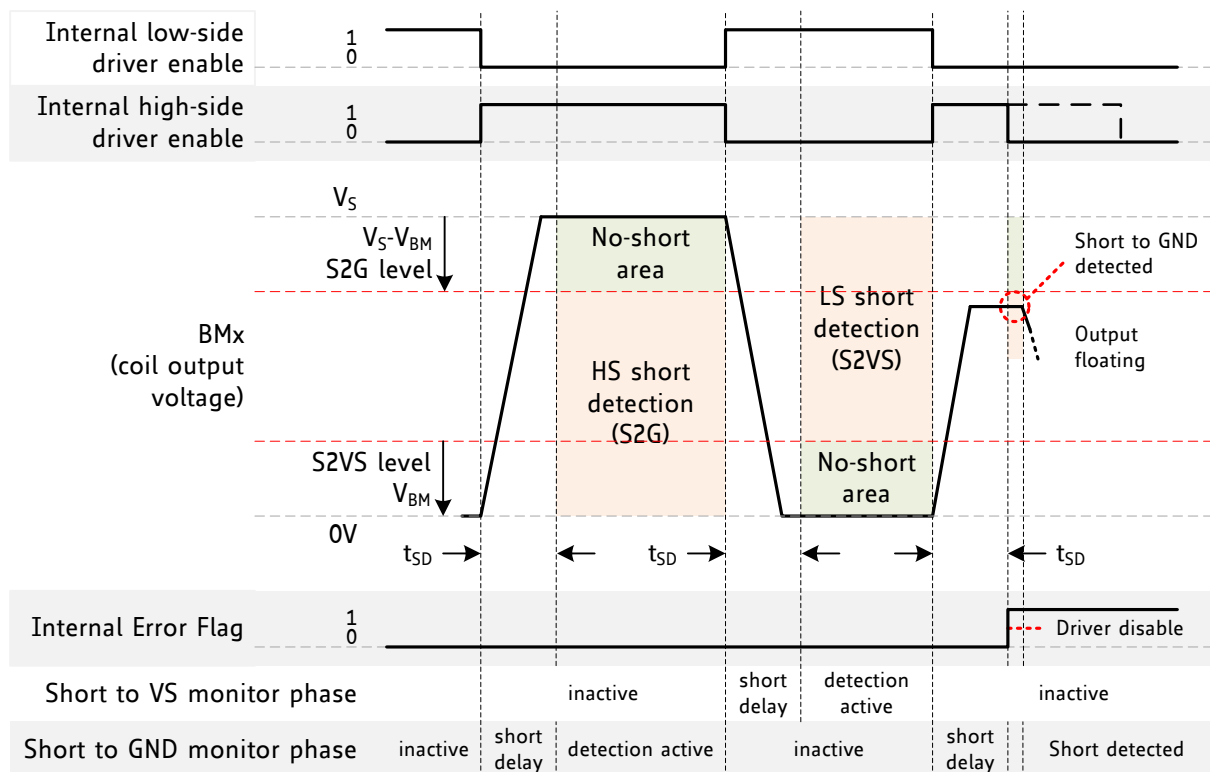


Figure 12.1 Short detection timing

As the low-side short detection includes the sense resistor, it is sensitive to the actual sense resistor and provides a good precision of overcurrent detection. This way, it will safely cover most overcurrent conditions.

Status flag	Description	Range	Comment
SHORTA	The SHORT bits identify a short condition on coil A or coil B persisting for multiple chopper cycles. The bits are cleared when the MOSFETs are disabled. Detailed warning for high-side or low-side FETs is available in S2VSA and S2VSB (low-side short detector) and S2GA and S2GB (high-side short detector)	0 / 1	0: No short detected 1: Short detected
SHORTB			
S2VSA			
S2VSB			
S2GA			
S2GB			

Hint

Once a short condition is safely detected, the corresponding driver bridge (A or B) becomes switched off, and the *s2ga* or *s2gb* flag, respectively *s2vsa* or *s2vsb* becomes set. To restart the motor, disable and re-enable the driver.

Attention

Short protection cannot protect the system and the power stages for all possible short events, as a short event is rather undefined, and a complex network of external components may be involved. Therefore, short circuits should basically be avoided.

12.2 Open-Load Detection

Interrupted cables are a common cause for systems failing, e.g., when connectors are not firmly plugged. The TMC2660 detects open load conditions by checking if it can reach the desired motor coil current. This way, also undervoltage conditions, high motor velocity settings or short and overtemperature conditions may cause triggering of the open load flag, and inform the user, that motor torque may suffer. In motor stand still, open load cannot be measured, as the coils might eventually have zero current.

Hint

Open load detection is provided for system debugging. Open load detection does not necessarily indicate a malfunction of the driver.

To safely detect an interrupted coil connection, check the open load flags following a motion of minimum four times the selected microstep resolution into a single direction using low or nominal motor velocity operation, only. However, the *ola* and *olb* flags have just informative character and do not cause any action of the driver.

False indication may result after changing motion direction or with full- or halfstep operation. Use microstepping for a safe result.

The open-load detection status is indicated by two bits:

Status flag	Description	Range	Comment
OLA	These bits indicate an open-load condition on coil A and coil B. The flags become set, if no chopper event has happened during the last period with constant coil polarity. The flag is not updated with too low actual coil current below 1/16 of maximum setting. It is a pure indicator. No action is taken depending on these flags.	0 / 1	0: No open-load detected 1: Open-load detected
OLB			

12.3 Temperature Sensors

The driver integrates a four-level temperature sensor (100°C pre-warning, 120°C overtemperature release and selectable 136°C / 150°C thermal shutdown) for diagnostics and for protection of the IC and the integrated MOSFETs as well as for adjacent components against excess heat. Choose the overtemperature level to safely cover error conditions like missing heat convection. Heat is mainly generated by the power MOSFETs, and, at increased voltage, by the internal voltage regulators. For many applications, already the overtemperature pre-warning will indicate an abnormal operation situation and can be used to initiate user warning or power reduction measures like motor current reduction. The thermal shutdown is just an emergency measure and temperature rising to the shutdown level should be prevented by design.

Hint for TMC2660C, only

After triggering the overtemperature sensor (ot flag), the driver remains switched off until the system temperature falls below the overtemperature release level (120°C) to avoid continuous heating to the shutdown level. Cool down typically needs a few 100ms to 1 second.

The high-side P-channel gate drivers have a temperature dependency which can be compensated to some extent by increasing the gate driver current when the warning temperature threshold is reached. The chip automatically corrects for the temperature dependency above the warning temperature when the temperature-compensated modes of SLPH is used. In these modes, the gate driver current is increased by one step when the temperature warning threshold is reached.

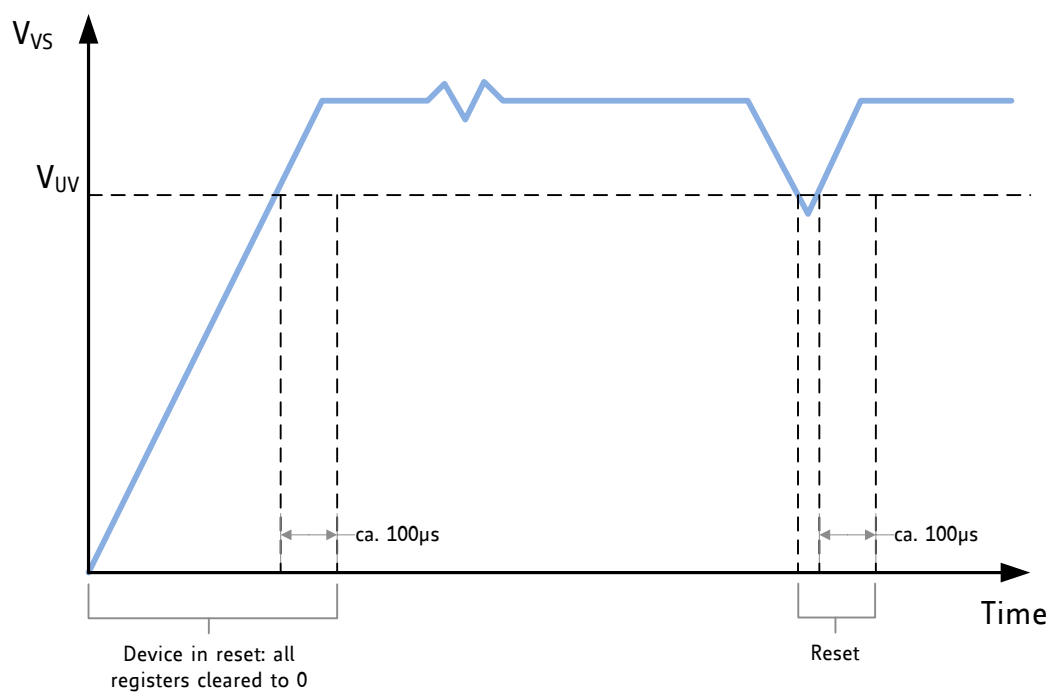
Status	Description	Range	Comment
OTPW	Overtemperature warning. This bit indicates whether the warning threshold is reached. Software can react to this setting by reducing current.	0 / 1	1: temperature prewarning level reached
OT	Overtemperature shutdown. This bit indicates whether the shutdown threshold has been reached and the driver has been disabled.	0 / 1	1: driver shut down due to over-temperature
OTSENS (TMC2660C only)	Program overtemperature threshold to a reduced level to give better protection for the power stage and other components. The driver re-enables when the temperature falls below 120°C.	0 / 1	0: 150°C 1: 136°C

12.4 Undervoltage Detection

The undervoltage detector monitors both the internal logic supply voltage and the supply voltage. It prevents operation of the chip when the MOSFETs cannot be guaranteed to operate properly because the gate drive voltage is too low. It also initializes the chip at power up.

In undervoltage conditions, the logic control block becomes reset and the driver is disabled. All MOSFETs are switched off. All internal registers are reset to zero. Software should additionally monitor the supply voltage to detect an undervoltage condition. If software cannot measure the supply voltage, an undervoltage condition can be detected by reading out the SE current value. Following a reset due to undervoltage occurs, the CS parameter is cleared, which is reflected in an SE status of 0 in the read response.

Status	Description	Range	Comment
EN_PFD (TMC2660C only)	Set this bit to allow operation with a 5V only supply. This bit additionally controls resonance dampening of the motor.	0 / 1	Undervoltage level 0: <7V 1: <4.5V

**Figure 12.2 Undervoltage reset timing****Note**

Be sure to operate the IC significantly above the undervoltage threshold to ensure reliable operation! Check for SE reading back as zero to detect an undervoltage event.

13 Power Supply Sequencing

The TMC2660 generates its own 5V supply for all internal operations. The internal reset of the chip is derived from the supply voltage regulator to ensure a clean start-up of the device after power up. During start up, the SPI unit is in reset and cannot be addressed. All registers become cleared.

VCC_IO limits the voltage allowable on the inputs and outputs and is used for driving the outputs. It is included in undervoltage detection and reset (C-type, only). Therefore, the startup sequence of the VCC_IO power supply with respect to VS is not important. VCC_IO may start up before or after VS.

14 System Clock

The system clock is the timing reference for all functions. The internal system clock frequency for all operations is nominally 15MHz. An external clock of 8MHz to 20MHz can be supplied for more exact timing, especially when using CoolStep and StallGuard2.

USING THE INTERNAL CLOCK

To use the on-chip oscillator of the TMC2660, tie CLK to GND near the chip. The actual on-chip oscillator clock frequency can be determined by measuring the delay time between the last step and assertion of the STST (standstill) status bit, which is 2^{20} clocks. There is some delay in reading the STST bit through the SPI interface, but it is easily possible to measure the oscillator frequency within 1%. Chopper timing parameters can then be corrected using this measurement, because the oscillator is relatively stable over a wide range of environmental temperatures.

Hint

In case well defined precise motor chopper operation are desired, it is supposed to work with an external clock source.

USING EXTERNAL CLOCK

An external clock frequency of up to 20MHz can be supplied. It is recommended to use an external clock frequency between 10MHz and 16MHz for best performance. Pay attention to using a near 50% duty-cycle. The external clock is enabled, and the on-chip oscillator is disabled with the first logic high driven on the CLK input.

Attention:

Never leave the external clock input floating. It is not allowed to remain within the transition region (between valid low and high levels), as spurious clock signals might lead to short impulses and can corrupt internal logic state. Provide an external pull-down resistor, in case the driver pin (i.e., microcontroller output) does not provide a safe level directly after power up. If the external clock is suspended or disabled after the internal oscillator has been disabled, the chip will not operate. Be careful to switch off the power MOSFETs (by driving the ENN input high or setting the TOFF parameter to 0) before switching off the clock, because otherwise the chopper would stop, and the motor current level could rise uncontrolled. If the short to GND detection is enabled, it stays active even without clock.

The TMC2660C provides a clock failover option to switch back to internal clock, in case of the external clock failing. The function is coupled to enabling short to VS protection (EN_S2VS).

Status	Description	Range	Comment
EN_S2VS (TMC2660C only)	Set this bit to enable protection against a failing external CLK source. If set, the IC switches back to external clock after 32 to 48 internal clock cycles. At the same time, this bit controls the higher side short detector sensitivity	0 / 1	Undervoltage level 0: Stopping clock will stop the IC. 1: CLK fail safe protection

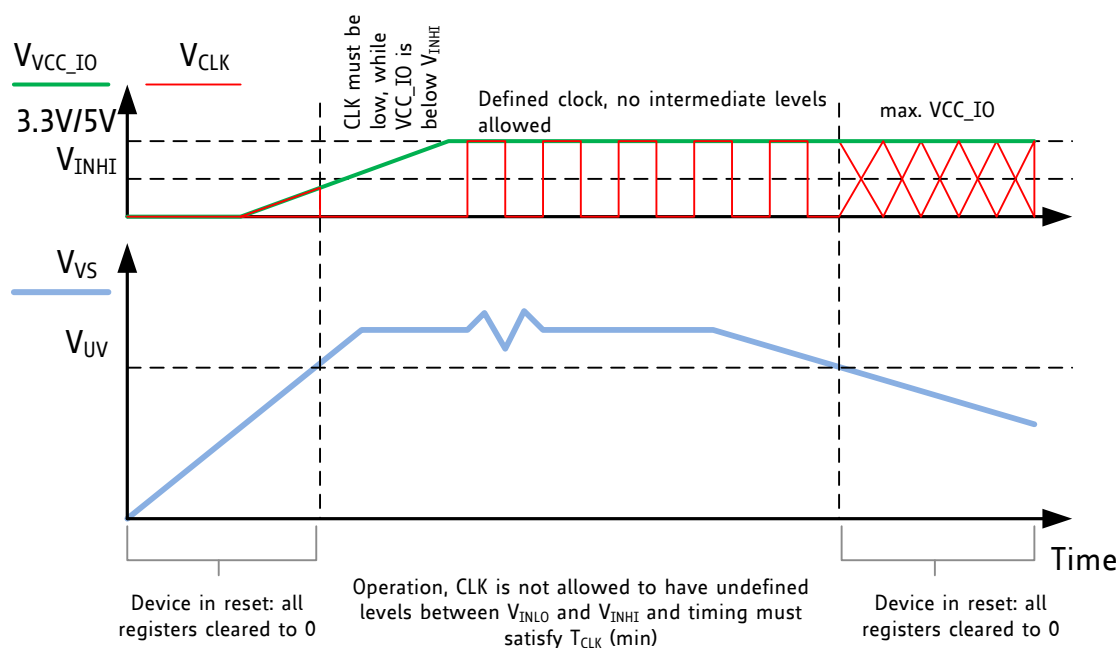


Figure 14.1 Start-up requirements of CLK input

14.1 Frequency Selection

A higher frequency allows faster step rates, faster SPI operation, and higher chopper frequencies. On the other hand, it may cause more electromagnetic emission and more power dissipation in the digital logic. Generally, a system clock frequency of 10MHz to 16MHz is best for most applications, unless the motor is to operate at the highest velocities. If the application can tolerate reduced motor velocity and increased chopper noise, a clock frequency of 4MHz to 10MHz should be considered.

CLK frequency	Comment
internal	13-16MHz, typical. Tie clock input firmly to GND. See electrical characteristics for limits. The internal clock is sufficient, unless a good reproducibility of StallGuard values is desired.
4-10 MHz	Lower range sufficient for higher inductance motors
10-16 MHz	Recommended for best results
16-20 MHz	Option in case a lower frequency is not available Ensure 40-60% duty cycle

15 Compatibility to non-C-type

The TMC2660C is downward compatible to the TMC2660. It is designed for compatibility to existing applications, while offering several enhancements and options for new designs.

ENHANCEMENTS IN C-TYPE

- More silent chopper operation, silent motor at low motor velocity.
- VCC_IO is monitored by internal reset circuitry to ensure clean power-up and power-down.
- Overtemperature shutdown has added hysteresis. After exceeding the temperature threshold, the driver stage remains off, until temperature falls below the 120°C pre-warning threshold.
- Reduced power dissipation of internal circuitry, leading to lower heat-up.
- CMOS input levels are increased with 5V VCC_IO level to increase noise rejection.
- Reduced filtering time for STEP & DIR inputs to reduce the risk of missed step pulses.

OPTIONAL ENHANCEMENTS IN C-TYPE (ENABLE VIA CONFIGURATION BITS)

- Clock fail-safe option. In case the external clock fails, the IC defaults back to internal clock. This avoids damage to motor and driver stage. (Enable: EN_S2VS, together with short protection)
- Low-Side short detection (Short to VS) for improved robustness in case of mis-wiring or overcurrent, e.g. due to wrongly attached motor. (Enable: EN_S2VS)
- Optional higher sensitivity high-side protection for improved protection. (Enable: SHRTSENS)
- Optional lower overtemperature threshold of 136°C for more sensitive protection of the power driver. Enable: OTSENSE
- Option to work down to 5V supply voltage and to add resonance dampening. (Enable: EN_PFD)

The additional settings for C-types are highlighted in this document using green background.

The major differences to be considered when migrating an existing application from the TMC2660 to TMC2660C, are summarized in the following table:

Item	Change in C-type	Impact
More silent chopper in C-type	Internal noise sources have been reduced. Noise caused by internal wiring voltage drop makes up for up to 10mV for non-C-devices. This reflects in a slightly lower sense input threshold voltage.	More silent motor, especially with low velocity. C-type effective motor current about 2-3% higher.
Undervoltage monitoring of VCC_IO	IO voltage VCC_IO is monitored by the internal reset circuitry. The device becomes reset upon VCC_IO undervoltage.	Increased VCC_IO supply current (50µA), device resets upon VCC_IO undervoltage.
Hysteresis for overtemperature detector	Following overtemperature detection at 136°C or 150°C, the driver stage remains switched off until the IC cools down below 120°C.	Longer shutdown time upon overtemperature detection, safe detection by interface.
Input levels depend on VCC_IO voltage	Non-C-type input level detectors are supplied by internal 5V supply, leading to fixed 0.8V and 2.4V levels. C-types use VCC_IO supplied CMOS Schmitt-Trigger inputs instead.	Higher positive logic level for 5V VCC_IO supply. No change with 3.3V VCC_IO supply. Check levels in 5V systems.
Spike filtering time on STEP/DIR input	The filtering has been improved, while reducing the filtering time in order to reduce the risk of missing short step pulses.	Potential impact in systems with high noise level on STEP & DIR input pins.
CLK input filter	C-type provides better filtering against short pulses (10ns typ.).	More robust against reflections on CLK. Check timing in case of SPI rate at $\frac{1}{2} f_{CLK}$. Invert CLK signal if required.

16 Driver Protection and EME Circuitry

Some applications have to cope with ESD events caused by motor operation or external influence. Despite ESD circuitry within the driver chips, ESD events occurring during operation can cause a reset or even a destruction of the motor driver, depending on their energy. Especially plastic housings and belt drive systems tend to cause ESD events of several kV. It is best practice to avoid ESD events by attaching all conductive parts, especially the motors themselves to PCB ground, or to apply electrically conductive plastic parts. In addition, the driver can be protected up to a certain degree against ESD events or live plugging / pulling the motor, which also causes high voltages and high currents into the motor connector terminals. A simple scheme uses capacitors at the driver outputs to reduce the dV/dt caused by ESD events. Larger capacitors will bring more benefit concerning ESD suppression, but cause additional current flow in each chopper cycle, and thus increase driver power dissipation, especially at high supply voltages. The values shown are example values – they might be varied between 100pF and 1nF. The capacitors also dampen high frequency noise injected from digital parts of the application PCB circuitry and thus reduce electromagnetic emission. A more elaborate scheme uses LC filters to de-couple the driver outputs from the motor connector. Varistors in between of the coil terminals eliminate coil overvoltage caused by live plugging. Optionally protect all outputs by a varistor against ESD voltage.

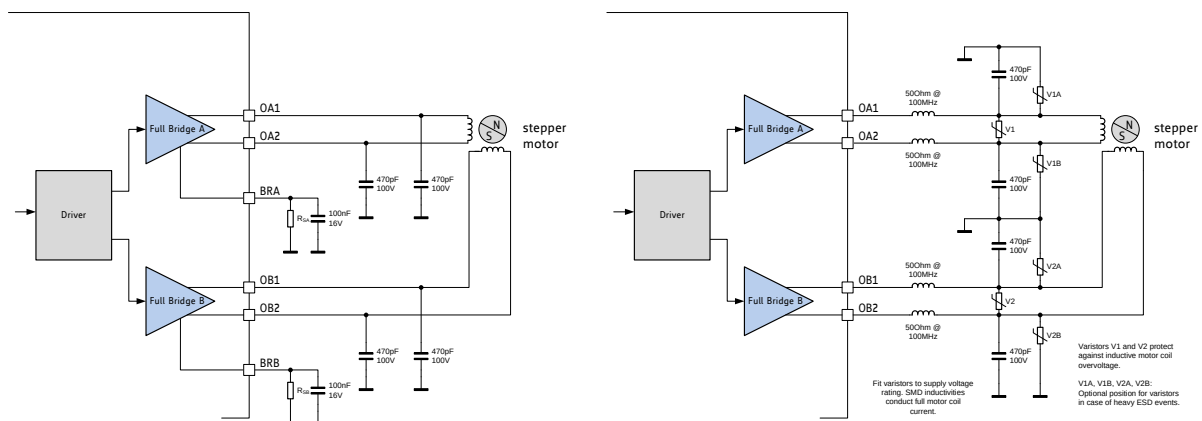


Figure 16.1 Simple ESD enhancement and more elaborate motor output protection

17 Layout Considerations

The PCB layout is critical to good performance, because the environment includes both high-sensitivity analog signals and high-current motor drive signals.

17.1 Sense Resistors

The sense resistors are susceptible to ground differences and ground ripple voltage, as the microstep current steps result in voltages down to 0.5mV. No current other than the sense resistor currents should flow through their connections to ground. Place the sense resistors close to chip with one or more vias to the ground plane for each sense resistor.

The sense resistor layout is also sensitive to coupling between the axes. The two sense resistors should not share a common ground connection trace or vias, because PCB traces have some resistance.

17.2 Power MOSFET Outputs

The OA and OB dual pin outputs on the TMC2660 are directly connected electrically and thermally to the drain of the MOSFETs of the power stage. A symmetrical, thermally optimized layout is required to ensure proper heat dissipation of all MOSFETs into the PCB. Use thick traces and areas for vertical heat transfer into the GND plane and enough vias for the motor outputs.

The printed circuit board should have a solid ground plane spreading heat into the board and providing for a stable GND reference. All signals of the TMC2660 are referenced to GND. Directly connect all GND pins to a common ground area to avoid voltage differences disturbing the operation.

The switching motor coil outputs have a high dV/dt , so stray capacitive coupling into high-impedance signals can occur, if the motor traces are parallel to other traces over long distances.

17.3 Power Supply Pins

Both, the VSA and VSB pins, as well as the BRA and BRB pins conduct the full motor current for a limited amount of time during each chopper cycle. Due to the resistance of bond wires connected to these pins, the pins heat up. Therefore, it is essential for current capability above 2A RMS to use a wide PCB trace for cooling and to avoid additional heat up of the pins caused by PCB trace resistance. This is simplified by also contacting the N.C. pins located next to VSA and VSB to the supply voltage. Failure to do so might affect reliability; despite heat-up of bond wires might not be visible with a thermal camera.

17.4 Power Filtering

The 470nF ceramic filtering capacitor on 5VOUT should be placed as close as possible to the 5VOUT pin, with its GND return going directly to the nearest GND pin. Use as short and as thick connections as possible. A 100nF filtering capacitor should be placed as close as possible from the VS pin to the ground plane. The motor supply pins, VSA and VSB, should be decoupled with an electrolytic (>100μF recommended) capacitor and a ceramic capacitor, placed close to the device. Also place minimum 100nF capacitor for VCC_IO close to the supply pin.

17.5 Layout Example

EXAMPLE FOR UP TO 2.8A RMS

Here, an example for a layout with small board size ($\approx 20\text{cm}^2$) is shown. See also the evaluation board.

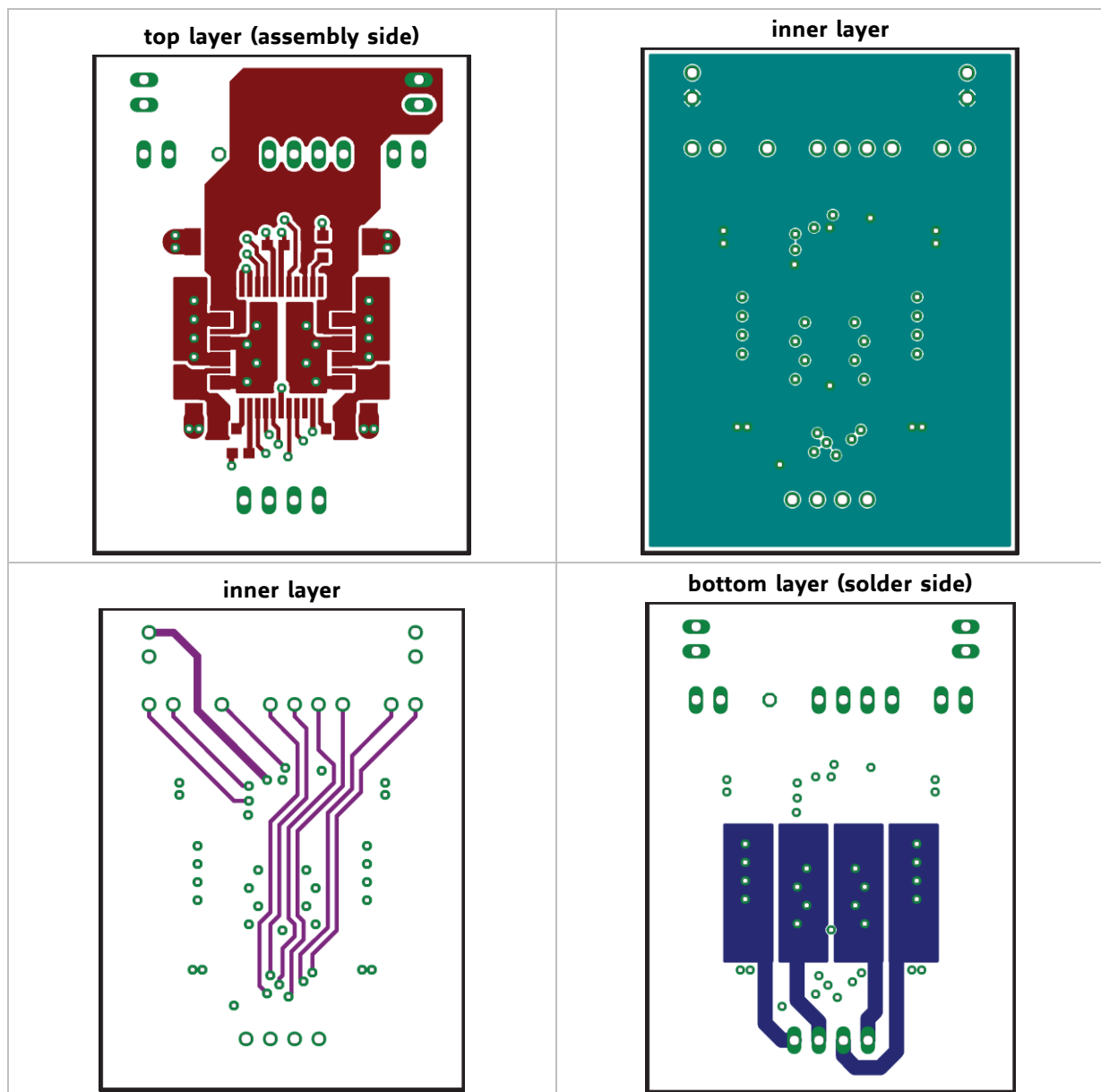
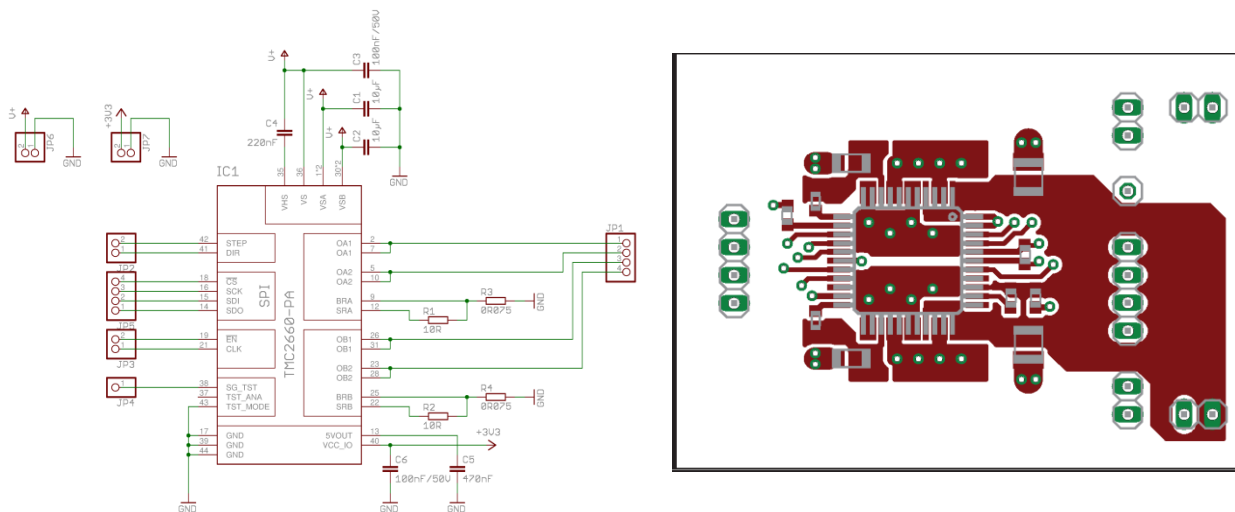


Figure 17.1 Layout example for TMC2660

18 Absolute Maximum Ratings

The maximum ratings may not be exceeded under any circumstances. Operating the circuit at or near more than one maximum rating at a time for extended periods shall be avoided by application design.

Parameter	Symbol	Min	Max	Unit
Supply voltage	V_{VS}	-0.5	30	V
Supply voltage when disabled (ENN V_{VIO}) with $I_{OXX}=0$	V_{VSDIS}		60	V
Logic supply voltage	V_{VCC}	-0.5	6.0	V
I/O supply voltage	V_{VIO}	-0.5	6.0	V
Logic input voltage	V_I	-0.5	$V_{VIO}+0.5$	V
Analog input voltage	V_{IA}	-0.5	$V_{CC}+0.5$	V
Relative high-side gate driver voltage ($V_{VM} - V_{HS}$)	V_{HSVM}	-0.5	15	V
Maximum current to/from digital pins and analog low voltage I/Os	I_{IO}		+/-10	mA
Non-destructive short time peak current into input/output pins	I_{IO}		500	mA
Bridge output peak current (10µs pulse)	I_{OP}		+/-7	A
Output current, RMS per coil, $T_A \leq 50^\circ\text{C}$ 20cm ² board with sample layout, $\leq 40\text{kHz}$ chopper, fastest slope	running >4 fullsteps/s	I_{OC}	2.0	A
	duty cycle 2s on 6s off		2.5	
	standstill, single coil on (halfstep position) *)		2.2	
Output current, RMS per coil, $T_A \leq 50^\circ\text{C}$ 50cm ² board with sample layout $\leq 40\text{kHz}$ chopper, fastest slope	running >4 fullsteps/s	I_{OC}	2.2	A
	duty cycle 2s on 6s off		2.8	
	standstill, single coil on (halfstep position) *)		2.4	
Output current, RMS per coil, $T_A \leq 85^\circ\text{C}$ 20cm ² board with sample layout, $\leq 40\text{kHz}$ chopper, fastest slope	running >4 fullsteps/s	I_{OC}	1.6	A
	duty cycle 2s on 6s off		2.0	
	standstill, single coil on (halfstep position) *)		1.8	
Output current, RMS per coil, $T_A \leq 85^\circ\text{C}$ 50cm ² board with sample layout $\leq 40\text{kHz}$ chopper, fastest slope	running >4 fullsteps/s	I_{OC}	1.8	A
	duty cycle 2s on 6s off		2.3	
	standstill, single coil on (halfstep position) *)		2.0	
5V regulator output current	I_{SVOUT}		50	mA
5V regulator peak power dissipation ($V_{VM}-5V$) * I_{SVOUT}	P_{SVOUT}		1	W
Junction temperature	T_J	-50	150	°C
Storage temperature	T_{STG}	-55	150	°C
ESD-Protection (Human body model, HBM), in application	V_{ESDAP}		2	kV
ESD-Protection (Human body model, HBM), device handling	V_{ESDDH}		300	V

*) The standstill specification refers to a stepper motor stopped at a high current. Normally, standstill current should be reduced to a value far below the run current to reduce motor heating.

19 Electrical Characteristics

19.1 Operational Range

Parameter	Symbol	Min	Max	Unit
Junction temperature	T_J	-40	125	°C
Supply voltage TMC2660C	V_{VS}	5	29	V
I/O supply voltage	V_{VIO}	3.00	5.25	V

19.2 DC and AC Specifications

DC characteristics contain the spread of values guaranteed within the specified supply voltage range unless otherwise specified. Typical values represent the average value of all parts measured at +25°C. Temperature variation also causes some values to stray. A device with typical values will not leave Min/Max range within the full temperature range.

Power Supply Current	DC Characteristics $V_{VS} = 24.0V$					
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Supply current, operating	I_{VS}	$f_{CLK}=16MHz$, 40kHz chopper		8		mA
Supply current, MOSFETs off	I_{VS}	$f_{CLK}=12MHz$		5		mA
Supply current, MOSFETs off, dependency on CLK frequency	I_{VS}	f_{CLK} variable <i>additional to I_{VS0}</i>		0.1		mA/ MHz
Static supply current	I_{VS0}	$f_{CLK}=0Hz$, digital inputs at +5V or GND		3.5	5.7	mA
Part of supply current NOT consumed from 5V supply	I_{VSHV}	MOSFETs off		1.2		mA
IO supply current	I_{VIO}	No load on outputs, inputs at V_{IO} or GND		50	100	µA

High-Side Voltage Regulator	DC-Characteristics $V_{VS} = 24.0V$					
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Output voltage ($V_{VS} - V_{HS}$)	V_{HSVS}	$I_{OUT} = 0mA$ $T_J = 25^\circ C$	9.3	10.0	10.8	V
Lower voltage for VHS regulator to activate	V_{VS}	VS rising, first time VHS goes up from 0V		12.5	13	V
Output resistance	R_{VHS}	Static load		50		Ω

Linear Regulator	DC Characteristics					
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Output voltage	V_{SVOUT}	$I_{SVOUT} = 10mA$ $T_J = 25^\circ C$	4.75	5.0	5.25	V
Output resistance	R_{SVOUT}	Static load			1	Ω
Deviation of output voltage over the full temperature range	$V_{SVOUT(DEV)}$	$I_{SVOUT} = 10mA$ $T_J = \text{full range}$		0	60	mV

Internal MOSFETs TMC2660C		DC Characteristics				
		$V_{VS} = V_{VSX} \geq 12.0V$, $V_{BRX} = 0V$				
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
N-channel MOSFET on resistance	R_{ONN}	$T_J = 25^{\circ}C$		63	76	m Ω
P-channel MOSFET on resistance	R_{ONP}	$T_J = 25^{\circ}C$		93	110	m Ω
N-channel MOSFET on resistance	R_{ONN}	$T_J = 150^{\circ}C$		110		m Ω
P-channel MOSFET on resistance	R_{ONP}	$T_J = 150^{\circ}C$		160		m Ω

Clock Oscillator and CLK Input		Timing Characteristics				
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Clock oscillator frequency	f_{CLKOSC}	$t_J = -50^{\circ}C$	10.0	13.5		MHz
Clock oscillator frequency	f_{CLKOSC}	$t_J = 50^{\circ}C$	10.8	14.3	17.5	MHz
Clock oscillator frequency	f_{CLKOSC}	$t_J = 150^{\circ}C$		14.5	18.0	MHz
External clock frequency (operating)	f_{CLK}	Typ. at 40%/60% dutycycle, Max at 50% dutycycle	4	10-16	20	MHz
External clock high / low level time	t_{CLK}		16			ns
External clock first pulse to trigger switching to external CLK	t_{CLKH} / t_{CLKL}		16			ns
External clock transition time	t_{TRCLK}	V_{INLO} to V_{INH1} or back			20	ns
External clock timeout detection in cycles of internal f_{CLKOSC}	$X_{timeout}$	External clock stuck at low or high	32		48	cycles f_{CLKOSC}

Detector Levels		DC Characteristics				
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
V _{VS} undervoltage threshold high	V _{UV}	EN_PFD=0	6.5	7	7.5	V
V _{VS} undervoltage threshold low	V _{UV}	EN_PFD=1	3.25	3.8	4.25	V
V _{VCC_IO} undervoltage threshold for RESET	V _{UV_VIO}	V _{VCC_IO} rising (delay typ. 10µs)	2.1	2.55	3.0	V
V _{VCC_IO} undervoltage detector hysteresis	V _{UV_VIOHYS} T			0.3		V
Short to GND detector threshold (high setting) (V _{VS} - V _{BMx})	V _{BMS2G}		1.2	1.7	2.3	V
Short to GND detector threshold (sensitive setting) (V _{VS} - V _{BMx})	V _{BMS2G}		0.7	1.0	1.3	V
Short to VS detector threshold (V _{BMx})	V _{BMS2VS}		1.3	1.5	1.8	V
Short to GND detector delay (low-side gate off detected to short detection)	t _{S2G}	TS2G=00	2.0	3.2	4.5	µs
		TS2G=10		1.6		µs
		TS2G=01		1.2		µs
		TS2G=11		0.8		µs
Overtemperature warning	t _{OTPW}		85	100	115	°C
Overtemperature release	t _{OTR}	Temperature falling		120		°C
Overtemperature shutdown lo	t _{OTL}	Temperature rising		136		°C
Overtemperature shutdown hi	t _{OTH}	Temperature rising	135	150	170	°C

Sense Resistor Voltage Levels		DC Characteristics				
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Sense input peak threshold voltage (low sensitivity)	V _{SRTRIPL}	VSENSE=0 Cx=248; Hyst.=0	310	323	340	mV
Sense input peak threshold voltage (high sensitivity)	V _{SRTRIPH}	VSENSE=1 Cx=248; Hyst.=0	155	173	190	mV

Digital Logic Levels		DC Characteristics				
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Input voltage low level ^{a)}	V _{INLO}		-0.3		0.3 V _{VIO}	V
Input voltage high level ^{a)}	V _{INHI}		0.7 V _{VIO}		V _{VIO} +0.3	V
Output voltage low level	V _{OUTLO}	I _{OUTLO} = 2mA			0.2	V
Output voltage high level	V _{OUTH}	I _{OUTH} = -2mA	V _{VIO} -0.2			V
Input leakage current	I _{I LEAK}		-10		10	µA
Digital pin capacitance	C			3.5		pF

Notes:

- a) Digital inputs left within or near the transition region substantially increase power supply current by drawing power from the internal 5V regulator. Make sure that digital inputs become driven near to 0V and up to the V_{VIO} I/O voltage. There are no on-chip pull-up or pull-down resistors on inputs.

19.3 Thermal Characteristics

Parameter	Symbol	Conditions	Typ	Unit
Thermal resistance bridge transistor junction to ambient, soldered to 4-layer 20cm ² PCB (or 20cm ² size per driver IC for multiple driver board)	R _{THA14}	one bridge chopping, fixed polarity	80	K/W
	R _{THA24}	two bridges chopping, fixed polarity	50	K/W
	R _{THA44}	Motor running	37	K/W
Thermal resistance bridge transistor junction to ambient, soldered to 4-layer 50cm ² PCB	R _{THA44a}	Motor running	28	K/W
Power dissipation in bridge MOSFETs (MOSFETs at 125°C) 24V, 30kHz chopper, fast slope	P _{BRIDGES}	2A RMS per coil	2.6	W
	P _{BRIDGES}	2.2A RMS per coil	3.2	W
	P _{BRIDGES}	2.8A RMS per coil	5.0	W
Additional for core	P _{CORE}	24V supply, 16MHz f _{CLK}	0.28	W

When operating the device near its current limits, ensure a good thermal design of the PCB layout to avoid overheating of the power integrated MOSFETs. Due to its multichip construction with individual heat transfer for each MOSFET of the power stage to the PCB using two pins, thermal characteristics depend on the layout symmetry. The actual thermal resistance also depends on the duty cycle and the die temperature. Use the thermal characteristics and the sample layout as a guideline for your own board layout. In case, the driver is to be operated at high current levels, special care should be taken to spread the heat generated by the driver power bridges efficiently within the PCB.

The worst-case thermal resistance occurs during motor stand still with the motor stopped in a half step position (one coil full current, other coil 0), as well as cyclic in slow motion below 4FS/s. Assume roughly 80°C/W, when there is only one bridge chopping. This is the worst-case scenario for heat-up. In stand still, with two bridges chopping at identical current (fullstep position), thermal resistance is reduced, because the power dissipation is distributed to more MOSFETs. Reduce stand still current to 68% or less, to compensate for both stand still scenarios. When the motor is running, calculate thermal resistance for the complete chip (all 8 MOSFETs working).

The MOSFET and bond wire temperature should not exceed 150°C, despite temperatures up to 200°C will not immediately destroy the devices. But the package plastics will apply strain onto the bond wires, so that cyclic, repetitive exposure to temperatures above 150°C may damage the electrical contacts and increase contact resistance and eventually lead to contract break. As the MOSFET temperatures cannot be monitored within the system, it is a good practice to react to the temperature pre-warning by reducing motor current, rather than relying on the overtemperature switch off.

Check MOSFET temperature under worst case conditions not to exceed 150°C using a thermal camera to validate your layout. Please carefully check your layout against the sample layout or the layout of the TMC2660-Evaluation board on the TRINAMIC website in order to ensure proper cooling of the IC!

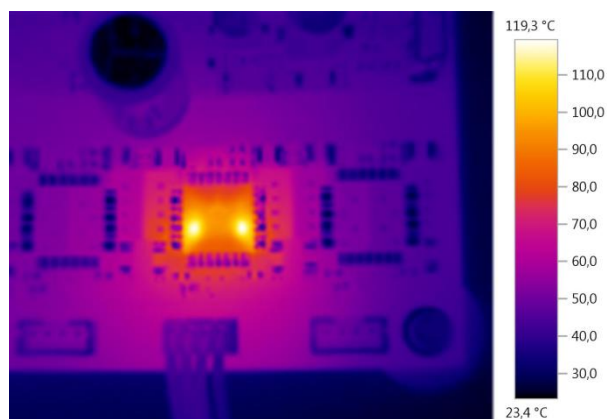


Figure 19.1 TMC2660 operating at 2.3A RMS (3.2A peak) on a 50cm² sized board

20 Package Mechanical Data

20.1 Dimensional Drawings

Attention: Drawings not to scale.

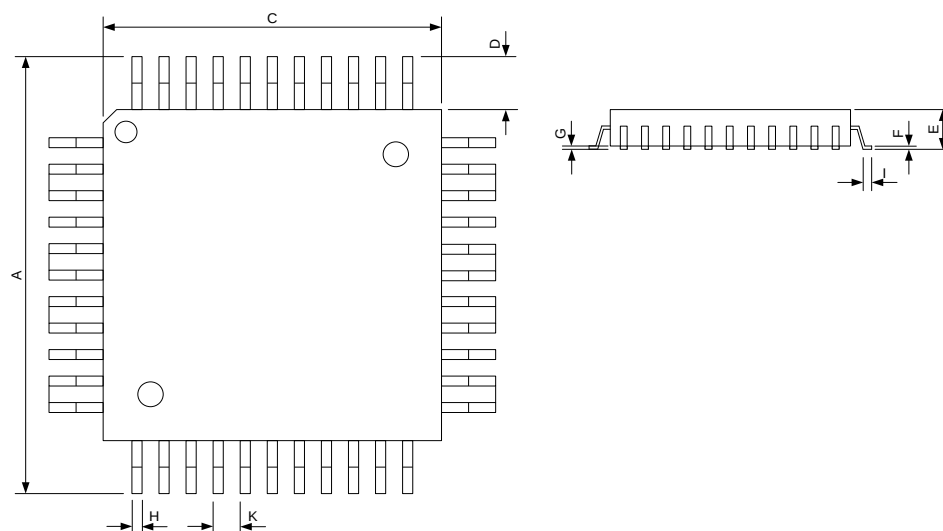


Figure 20.1 Dimensional drawings (PQFP44)

Parameter	Ref	Min	Nom	Max
Size over pins (X and Y)	A		12	
Body size (X and Y)	C		10	
Pin length	D		1	
Total thickness	E			1.6
Lead frame thickness	F	0.09		0.2
Stand off	G	0.05	0.10	0.15
Pin width	H	0.30		0.45
Flat lead length	I	0.45		0.75
Pitch	K		0.8	
Coplanarity	ccc			0.08

20.2 Package Code

Device	Package	Temperature range	Code/markings
TMC2660C	PQFP44 (RoHS)	-40° to +125°C	TMC2660C-PA / YYWW

WW=Week, YY=Year

21 Disclaimer

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22 ESD Sensitive Device

The TMC2660 is a ESD-sensitive CMOS device and sensitive to electrostatic discharge, due to discrete MOSFETs integrated into the package. Take special care to use adequate grounding of personnel and machines in manual handling. After soldering the device to the board, ESD requirements are more relaxed. Failure to do so can result in defects or decreased reliability.



Note: In a modern SMD manufacturing process, ESD voltages well below 100V are standard. A major source for ESD is hot-plugging the motor during operation. As the power MOSFETs are discrete devices, the device in fact is very rugged concerning any ESD event on the motor outputs. All other connections are typically protected due to external circuitry on the PCB.

23 Designed for Sustainability

Sustainable growth is one of the most important and urgent challenges today. We at Trinamic try to contribute by designing highly efficient IC products, to minimize energy consumption, ensure best customer experience and long-term satisfaction by smooth and silent run, while minimizing the demand for external resources, e.g., for power supply, cooling infrastructure, reduced motor size and magnet material by intelligent control interfaces and advanced algorithms.

Please help and design efficient and durable products made for a sustainable world.

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25 Revision History

Version	Date	Author BD – Bernhard Dwersteg	Description
1.00	2020-JUL-06	BD	New version for -C types. Non-C-type information only for reference
1.01	2020-AUG-11	BD	Corrected pinning table
1.02	2021-JUN-17	BD	Added EME example, minor fixes
1.03	2022-FEB-15	BD	Updated logo; corrected open load detection conditions; corrected upper limit for I_{V50} from 5 to 5.7mA.